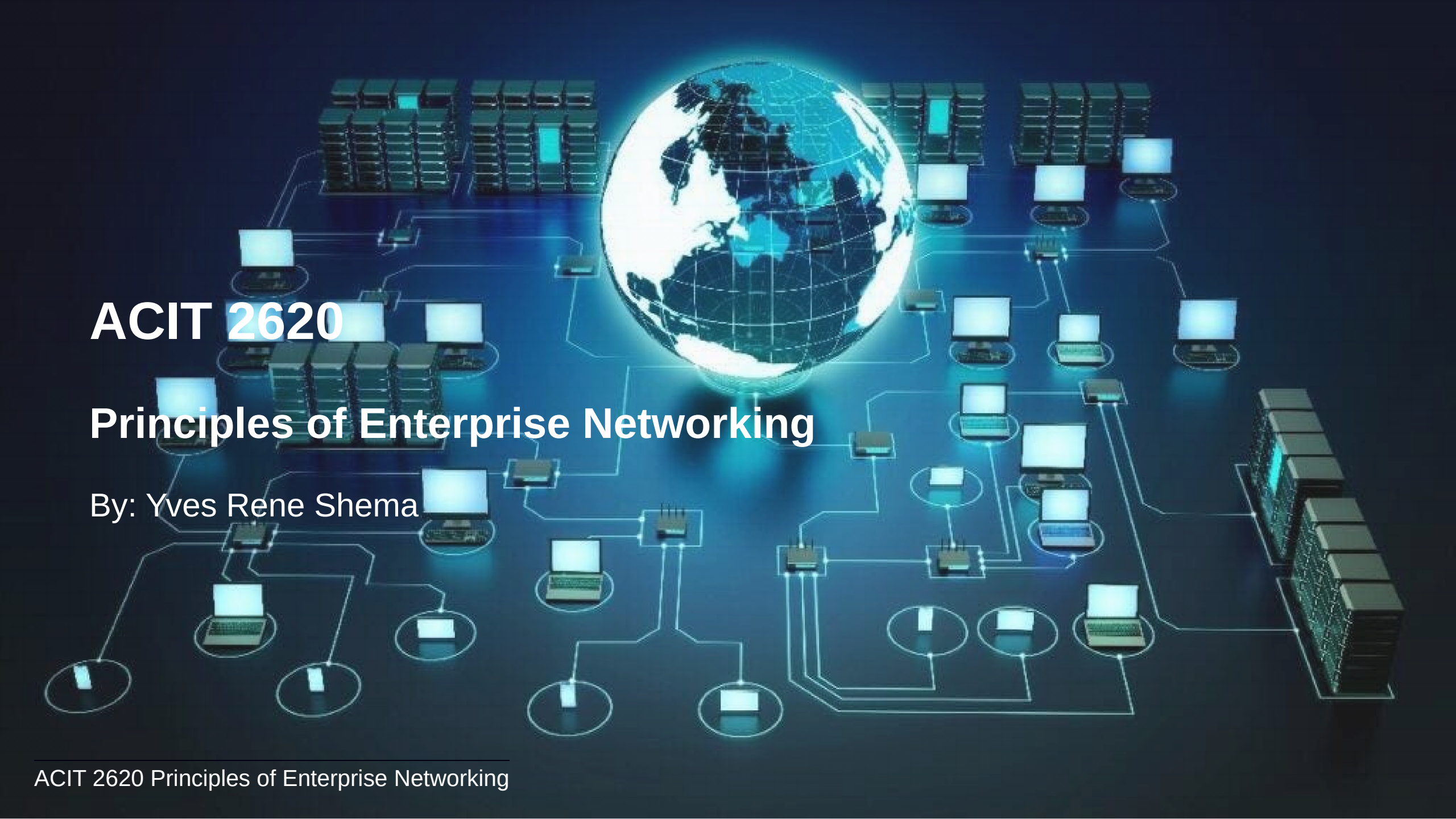


ACIT 2620

Principles of Enterprise Networking

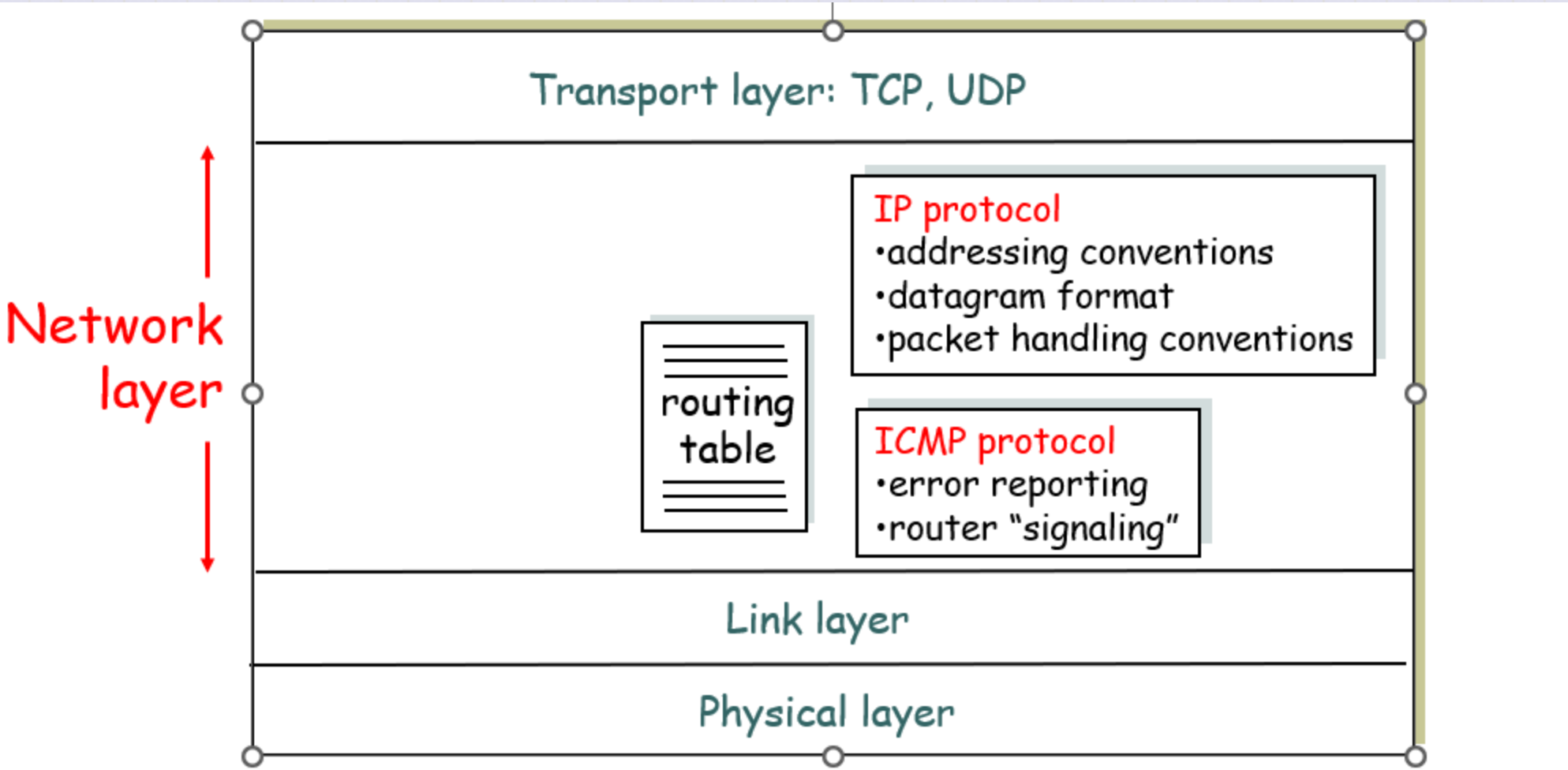
By: Yves Rene Shema

Internet Protocol Version 4

Network Layer

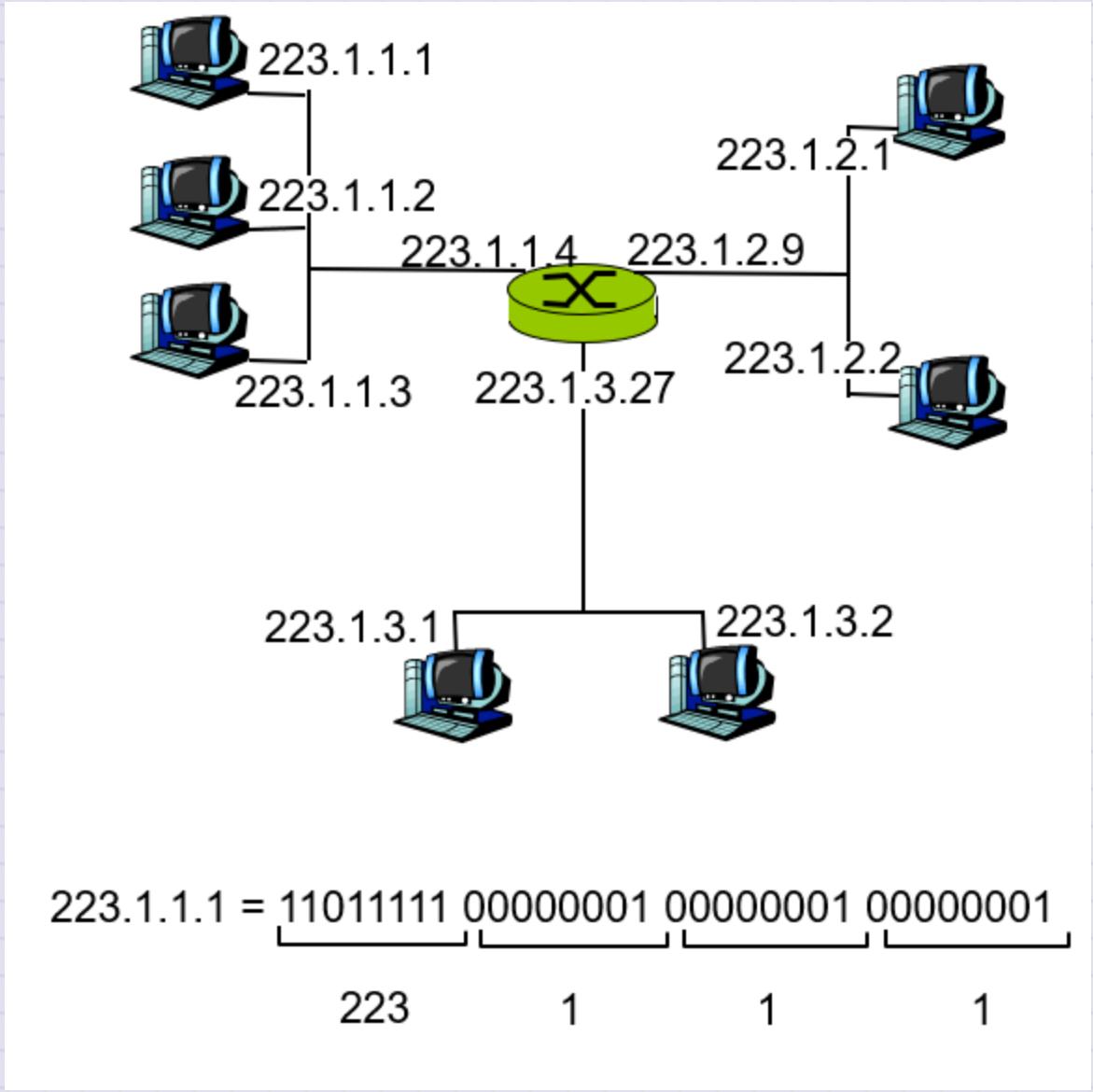
- **Goal**: move packets for source to destination
  - **Path Determination**:
    the calculation of the route taken by packets -> routing
  - **Forwarding**:
    The movement of a packet from one network to the next appropriate network

Network Layer functions



IP Addressing

- **IP address**: 32-bit identifier for host, router interface
- **interface**: connection between host, router and physical link
  - routers typically have multiple interfaces
  - host may have multiple interfaces
  - IP addresses associated with interface, not host, router


223.1.1.1
223.1.1.2
223.1.1.3
223.1.1.4
223.1.2.9
223.1.2.1
223.1.2.2
223.1.3.27
223.1.3.1
223.1.3.2
223.1.1.1 = 11011111 00000001 00000001 00000001
223 1 1 1

Components of an IP address

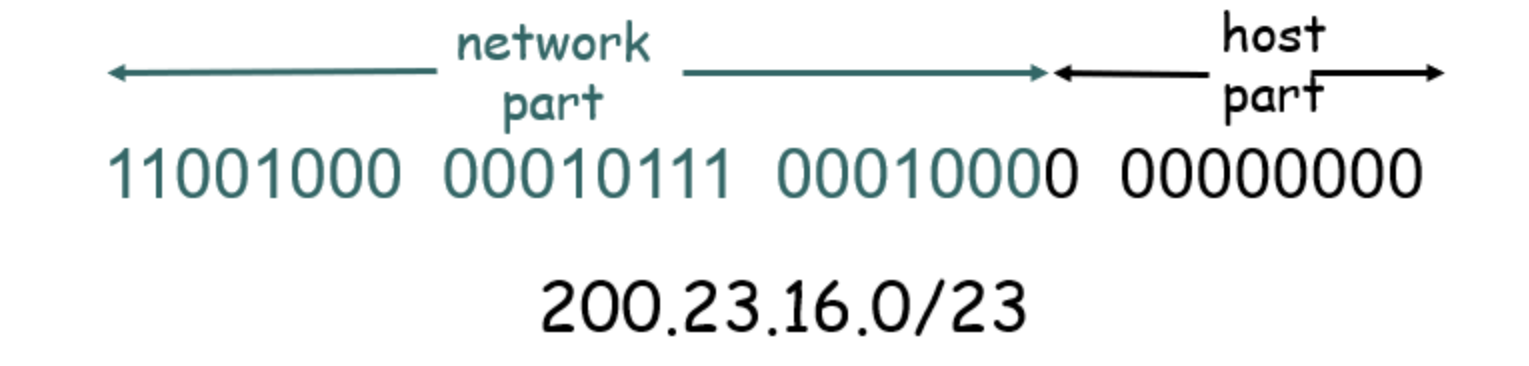
- network part (high order bits)
- host part (low order bits)

IP Network

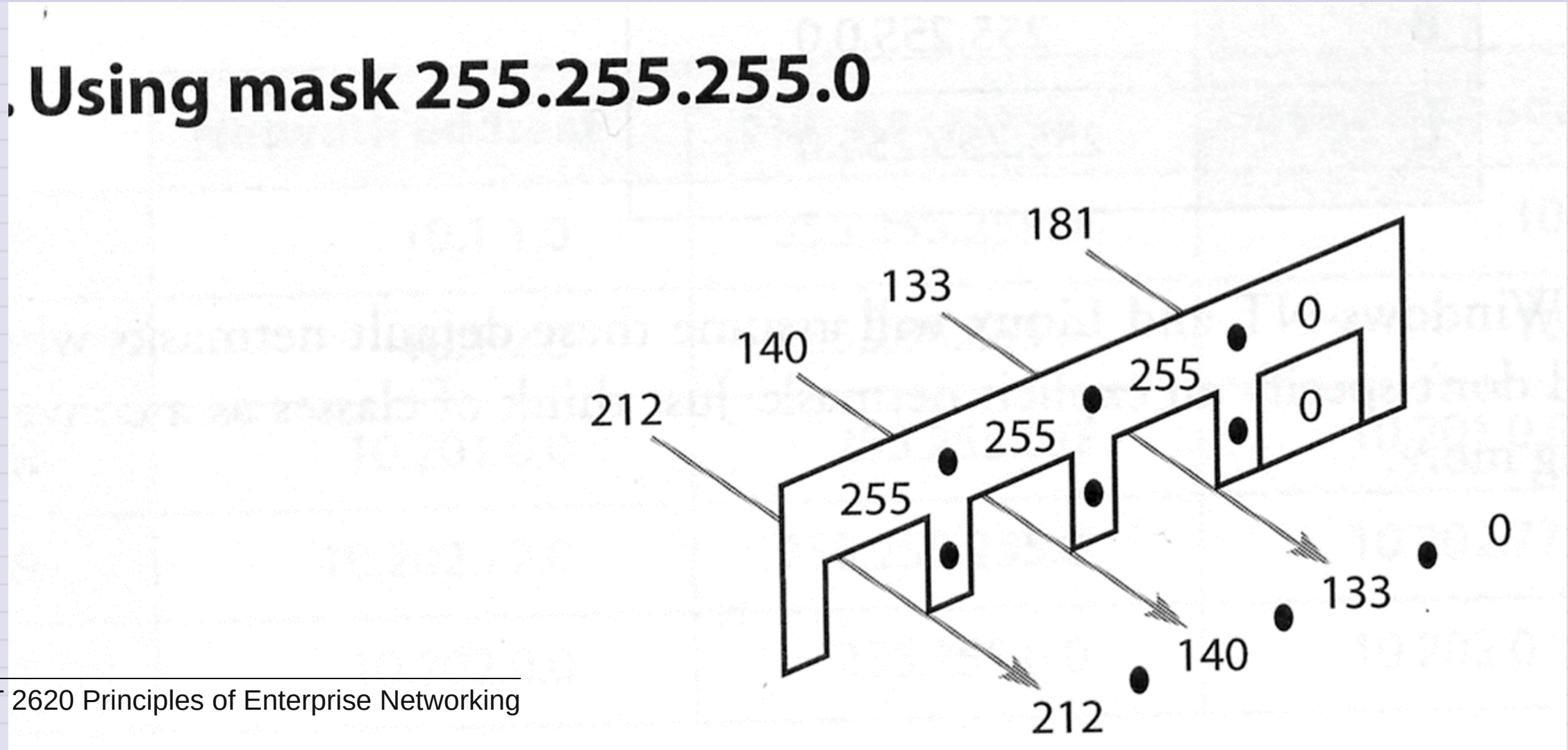
- device interfaces with same network part of IP address
- can physically reach each other without intervening router

IP range notations

- **CIDR**: Classless Inter Domain Routing
  - network portion of address of arbitrary length
  - address format: `a.b.c.d/x`, where `x` is # of bits in network portion of address
  - Also written as address + subnet mask



- CIDR using Subnet Mask
  - Usually written in dotted decimal notation reminiscent of an IP address



Special Addresses

- Private IP Addresses
  - 10.0.0.0/8
  - 172.16.0.0/12 (172.16 - 172.31)
  - 192.168.0.0/16

- Documentation IP Addresses
  - 192.0.2.0/24
  - 203.0.113.0/24
  - 198.51.100.0/24
- Self-Configured IP Addresses (often DHCP Failure)
  - 169.254.0.0/16

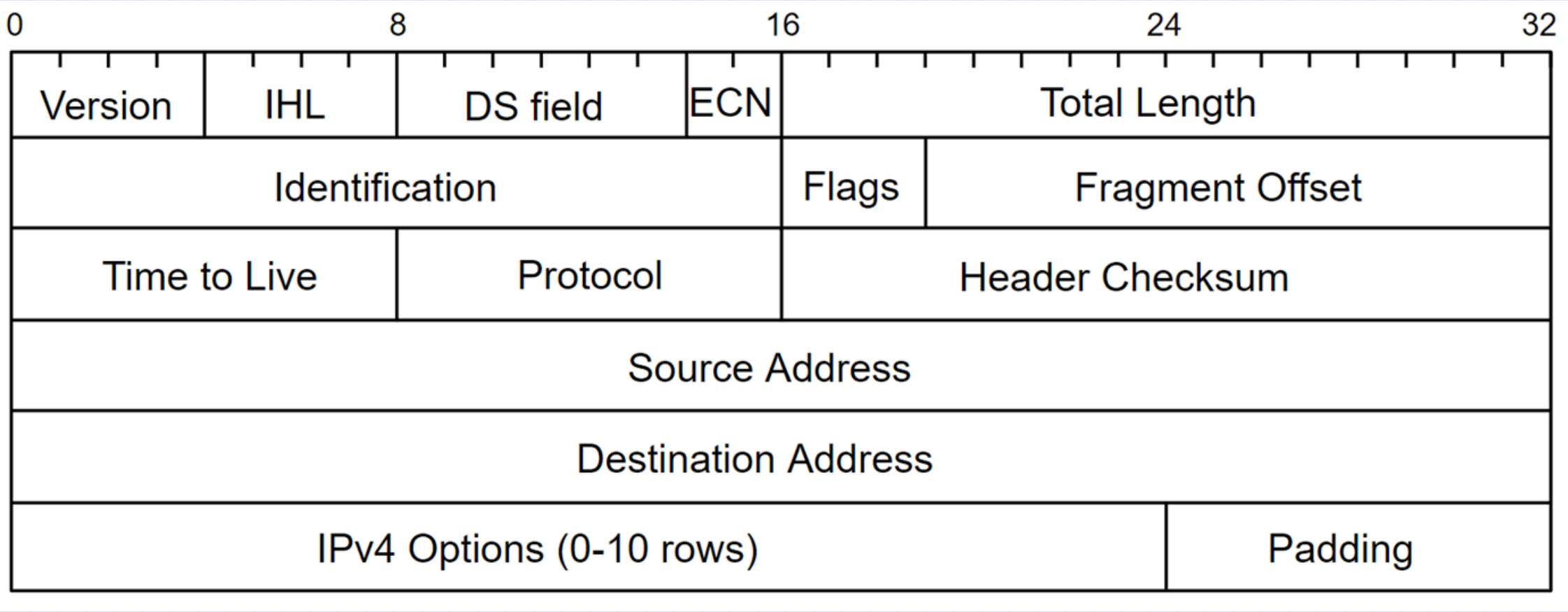
- Unknown Address
  - 0.0.0.0
- Loopback Address
  - 127.0.0.1 (actually, 127.0.0.0/8)
- Network Address (All host bits set to 0)
  - E.g: 192.168.1.0, Subnet Mask = 255.255.255.0

Broadcasting and Multicasting

- Limited Broadcast
  - 255.255.255.255
  - Transmitted only on local segment - not routed
- Network Broadcast Address
  - Network Address + All host bits set to one
    - Network Address = 192.168.1.x
    - Network Broadcast Address = 192.168.1.255

- Multicast Address
  - Lie within the 224.0.0.0 /4 network

IPv4 Header


0
8
16
24
32
Version
IHL
DS field
ECN
Total Length
Identification
Flags
Fragment Offset
Time to Live
Protocol
Header Checksum
Source Address
Destination Address
IPv4 Options (0-10 rows)
Padding